A. P. VIMALADHITHYA

Junior AI Engineer | Machine Learning Engineer

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PROFESSIONAL SUMMARY

Data Science Consultant with hands-on experience in Machine Learning, Deep Learning, Generative AI, NLP, and Predictive Analytics. Developed AI-driven solutions including a multi-agent Agriculture Advisor using **CrewAI** and **Gemini API** for crop recommendations and farming advisories based on location, soil, and weather data, and an ATS Resume Optimizer leveraging semantic similarity analysis, **NLP**, and LLM-based recommendations. Delivered a Credit Risk Prediction Model using **XGBoost**, achieving 96.36% accuracy and an AUC of 0.85. Proficient in Python, SQL, TensorFlow, PyTorch, Scikit-learn, Feature Engineering, Data Preprocessing, Exploratory Data Analysis (EDA), Model Evaluation, and Multi-Agent AI Systems. Seeking a Data Scientist or AI/ML Engineer role to develop scalable AI applications, intelligent automation solutions, and data-driven predictive models that deliver business impact.

TECHNICAL SKILLS

Programming & Scripting: Python, SQL, Java, Object-Oriented Programming (OOP)

AI & GenAI: Large Language Models (LLMs), Prompt Engineering, AI Agents, LangChain(Basic), LangGraph(Basic)

Backend: REST APIs, JSON, HTTP Methods (GET, POST, PUT, DELETE), Basic Node.js, API Integration

DevOps & Deployment: Docker, Git, GitHub, Streamlit, CI/CD (Basic)

Machine Learning: Regression, Classification, Clustering, XGBoost, LightGBM, Random Forest, Feature Engineering, Hyperparameter Tuning, Hypothesis Testing, SVM

Deep Learning & CV: PyTorch, TensorFlow, CNN, ANN, Transfer Learning

Data Analysis & Visualization: Advanced Excel (Pivot Tables, VLOOKUP, Dashboards), Statistical Analysis, Pandas, NumPy, Matplotlib, Seaborn, Power BI, Data Processing, Data Cleaning

Databases: MySQL, PostgreSQL, MongoDB

Cloud & AI: AWS EC2/S3 (basic), Text Preprocessing, Tokenization (basic), Natural Language Processing (NLP), Large Language Models (LLMs), CrewAI, Gemini API

WORK EXPERIENCE

Data Science Consultant | Rubixe, Bangalore *Sep 2025 – Apr 2026*

- Delivered Credit Risk modeling using XGBoost on 50K+ structured financial records, achieving 96.36% accuracy, AUC 0.85, and Gini score of 35.97.
- Built an Earthquake Damage Prediction classification model using XGBoost (best model), achieving Accuracy 71.3%, Precision 71.5%, Recall 71.3%, and F1-score 0.702 on seismic data.
- Performed data cleaning, preprocessing, feature engineering, and EDA on large-scale structured datasets using Python, Pandas, and NumPy.
- Optimized model performance through hyperparameter tuning and cross-validation; evaluated using Accuracy, AUC, Gini, Precision, Recall, and F1-score metrics.

Technologies: *Python, Pandas, NumPy, scikit-learn, XGBoost, LightGBM, Matplotlib, Seaborn*

KEY PROJECTS

AI-Powered Agriculture Advisor — CrewAI & AI-Powered Web Application *July 2026*

GitHub: github.com/vimaladhithya/AI-Powered-Agriculture-Advisor
Live Demo: AI-Powered Agriculture Advisor App

- Developed and deployed a multi-agent AI agricultural advisory system using CrewAI and Gemini LLM for data-driven farming recommendations..
- Designed Location, Soil, and Weather Agents to analyze geospatial, soil, and climate data for crop suitability assessment..
- Integrated agricultural datasets and real-time environmental data to generate personalized crop recommendations.
- Automated farming advisories for planting, irrigation, harvesting, and weather-risk management to support informed decision-making.

Technologies: *Python, CrewAI, Gemini API, Streamlit, Pandas, REST APIs*

ATS Score and Resume Optimizer — NLP & AI-Powered Web Application *June 2026*

GitHub: github.com/vimaladhithya/ATS-Score-and-Resume-Optimizer
Live Demo: ATS Resume Analyzer App

- Developed and deployed an ATS Resume Analyzer to evaluate resume compatibility with job descriptions and technical skill requirements.
- Implemented skill extraction, keyword matching, and ATS scoring to identify skill gaps and improve resume relevance.
- Integrated Sentence Transformers for semantic similarity analysis between resumes and job descriptions.
- Built an AI-powered resume optimization module using Gemini API to generate ATS-friendly, role-specific resume recommendations.

Technologies: *Python, Streamlit, NLTK, Sentence Transformers, Google Gemini API, ReportLab*

Credit Risk Prediction — (Client Project) *Feb – Mar 2026*

GitHub: github.com/vimaladhithya/credit-risk-prediction

- Designed a Credit Risk prediction model to classify loan defaulters using financial, demographic, and transactional data from 50K+ records.
- Conducted data cleaning, preprocessing, feature engineering, and EDA across three structured tables — Enquiry, Customer Account, and Demographic — to improve data quality and model readiness.
- Implemented and compared Random Forest, LightGBM, and XGBoost; applied hyperparameter tuning to optimize each model's performance.
- Achieved best performance with XGBoost: Accuracy 96.36%, Gini 35.97, AUC 0.85.

Technologies: *Python, Pandas, NumPy, scikit-learn, XGBoost, LightGBM, Matplotlib*

Earthquake Damage Prediction — Classification *Jan – Feb 2026*

GitHub: github.com/vimaladhithya/Earthquake_Prediction

- Built a multi-class classification model to predict earthquake damage levels from seismic feature data.
- Performed data preprocessing, feature selection, and EDA to identify key seismic attributes driving severity classification.
- Compared Random Forest, LightGBM, and XGBoost with hyperparameter tuning; XGBoost delivered the best results.
- Achieved Accuracy 71.3%, Precision 71.5%, Recall 71.3%, F1-score 0.702 with XGBoost as the final model.

Technologies: *Python, Pandas, NumPy, scikit-learn, XGBoost, LightGBM, Random Forest, Matplotlib*

PROFESSIONAL TRAINING

Certified Data Scientist Program — Datamites *Jul – Sep 2025*

- Completed intensive training in Machine Learning, Deep Learning, Statistics, Python, and SQL with hands-on end-to-end project work.
- Built and evaluated ML models on real-world datasets with practical experience in data preprocessing, feature engineering, and model evaluation.

CERTIFICATIONS

- Certified Data Scientist (CDS) — IABAC, 2026
- Data Science Foundation (DSF) — IABAC, 2026
- Certified Data Scientist — NASSCOM (Gold, 91%), 2026
- Certified Data Scientist Program — Machine Learning, Statistics & Deep Learning — Datamites, 2025
- SQL with PostgreSQL & pgAdmin — Udemy, 2025
- AI Tools Workshop — Be10x, 2025

EDUCATION

Bachelor of Engineering (Computer Science) *2021 – 2025*

Mepco Schlenk Engineering College, Sivakasi